

Whose Regulation Does the Model Know? Reliability of LLM-Recalled Regulatory Information for Public Environmental Management in 25 Countries

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Abstract

A municipal environmental manager asks a language model what the binding PM_{2.5} standard is in their country, and nothing in the answer says whether it matches the official register. We propose a protocol for that check and report what it finds. Answers are scored against official registers, by code wherever the correct answer is a published value and by a three-vendor judge panel elsewhere, across 25 countries, 14 deployment stacks and four languages (8,244 scored cells). None of the three fixes an agency can deploy without building anything works, and the most intuitive one harms. Asking in the country’s own language lowers accuracy by 5.0 percentage points (mixed model with a random intercept per country, $p = 6 \times 10^{-7}$; negative in 8 of nine countries) and makes the model 9.4 points less likely to return a register-checkable value at all. Declaring the official role is equivalent to declaring

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nothing within two percentage points. Regional instruction tuning at 7B does not improve on the base model it was tuned from. Accuracy is lowest on the task that carries administrative consequence, recalling the binding value (0.39 against 0.61 for synthesis and recommendation), and a third of the wrong answers cite a value from the country’s own official ladder. Global South countries return the register value 14.7 points less often than Global North ones with the country as the unit; the point estimates are largest where the answer is a national fact, but every per-task interval includes 1.

Keywords: algorithm auditing, geographic bias, multilingual bias, digital government, Global South

1. Introduction

Large language models have become working infrastructure of public administration within a few years. An OECD survey of core government functions documents generative-AI adoption across policymaking and service delivery, including assistants that help public-service advisers retrieve information and the sources behind it (OECD, 2025). Air pollution policy is a domain where this carries direct stakes for public health. Fine particulate matter is among the leading environmental risk factors for premature mortality worldwide (GBD 2019 Risk Factors Collaborators, 2020; World Health Organization, 2021), and the burden falls disproportionately on the Global South. When a public environmental manager asks a model about an ambient air quality standard, an attainment deadline, or the evidence behind an intervention, an inaccurate answer can propagate into a regulatory brief, a procurement decision, or an emergency response during a pollution episode.

This paper asks whether models answer such questions as reliably for Global South countries as for Global North ones, and whether the way a query is framed changes the answer. Both premises have support. Geographic factual bias is measurable: error rates are $1.5\times$ higher for Sub-Saharan African than for North American countries across 20 models (Moayeri et al., 2024), model ratings on subjective topics track socioeconomic conditions up to $\rho = 0.70$ (Manvi et al., 2024), and models systematically privilege statements about the Global North (Mirza et al., 2024). The linguistic asymmetry behind it is structural: the languages of most Global South countries occupy the under-served classes of the resource hierarchy (Joshi et al., 2020), and large Internet-derived corpora “overrepresent hegemonic viewpoints and encode biases potentially damaging to marginalized populations” (Bender et al., 2021). Decolonial scholarship reads that asymmetry as an uneven epistemic infrastructure reproduced by AI systems (Mohamed et al., 2020).

Three gaps organise the study. No benchmark targets regulatory information with official ground truth: existing evaluations use World Bank recall, subjective ratings, journalistic fact-checking, or everyday cultural knowledge, none of which measures the tasks a manager delegates, such as stating a binding standard, retrieving an officially measured concentration, summarising health-burden evidence, identifying legal instruments and executing agencies, or recommending an operational response. Persona framing is unexamined as an experimental lever, although practitioners routinely prefix a role and the vendors advise it (Section 2); whether it reduces the gap by steering toward regulation-compliant answers or amplifies it by inviting confident fabrication has not been tested in a pre-specified design. And the

corpus-representation mechanism is usually asserted rather than measured against confounders, and rarely separated into its two channels, the size of the language corpus (Xue et al., 2021; Abadji et al., 2022) and the breadth of country coverage, a distinction that matters here because the geographic gap is measured in English.

Objectives and scope

The study has three bounded objectives. The first is to establish a protocol for auditing regulatory information. Its object is agreement between a model’s answer and the value published in the official register, for tasks an environmental agency delegates; it does not measure model capability in general, reasoning quality, or usefulness for other purposes. The protocol is the contribution that transfers, and the domain is where we demonstrate it. The second is to test the mitigations available to an administration. A public body facing poor answers about its own country can declare the official role in the prompt, ask in the national language, or run a regionally fine-tuned open model, and we test all three as designed factors. The regional model we could run is a community fine-tune of a 7B base on Brazilian Portuguese, the kind an agency can download; the finding concerns that class, and the released protocol allows any other candidate to be tested. We do not test retrieval augmentation, fine-tuning, or human-in-the-loop verification, each of which requires engineering capacity the prompt-level fixes do not. All queries are answered from the model’s parametric memory, without web search or document retrieval, which is why the title speaks of recalled rather than retrieved information. The third objective is to locate where reliability differs and, as far as the design allows, why; we test the corpus-representation account and

claim no causal identification.

Throughout, the unit of analysis is the model as served, that is, weights, vendor and serving infrastructure reached through a pinned endpoint, because a public manager interacts with that stack and any measured gap is a property of it. Hypotheses, sampling and analysis plan were fixed before data collection (Nosek et al., 2018) for 15 countries and then extended to 25; the plan was never deposited publicly, so we do not claim pre-registration, and the study is reported as exploratory throughout.

Contributions of this work

1. *An auditing protocol for regulatory information.* The question an administration faces is whether the value a model returns matches the official register. We build five task types from the everyday work of an environmental agency (Table 1), each keyed to an official source, so that correctness is adjudicable rather than a matter of expert impression, and the protocol transfers to any regulated domain where the correct answer is published.
2. *The three localisation fixes agencies reach for, tested rather than assumed.* Persona framing, native-language prompting and a regionally tuned model are each a procurement or workflow decision. None of them works, and they fail differently: asking in the country’s own language lowers accuracy by 5.0 percentage points, most sharply for the language with the scarcest corpus; the role frame is equivalent to no frame within two percentage points; and the regionally tuned model recalls the binding value no more often than the base model it was tuned from.

3. *A design that separates who is poorly served from why, and reports where it cannot.* Countries are stratified on the UNCTAD developed/developing split, the Joshi linguistic-resource taxonomy, and development, so that geographic, linguistic and economic explanations can be told apart. Between countries, coverage and development are collinear; within countries, what predicts the country-specificity deficit at the country level is development, not the encyclopedic coverage the protocol pre-specified, and the two cannot be separated in this sample. We also document that the pre-specified corpus proxy assigns the English Wikipedia to every sample country with English as an official language, so that it tracks colonial history rather than corpus size.
4. *Ground truth adjudicated by code, and what that changes.* Rebuilding two task keys from official registers and moving range comparison into code cut the development gradient from $\rho = 0.51$ to $\rho = 0.36$ and more than doubled the native-language penalty on the same responses; moving the third register-valued task from a judge to code removed a confound between judge and tier. In this literature the ground-truth key, not the sample size, is the binding constraint.
5. *Open resources.* Protocol, prompts, ground-truth registries with their official sources, analysis code, and the deviation log are released under CC-BY-4.0 and MIT, with a one-command reproduction pipeline.

2. Related Work

Prior work runs along four threads: the measurement of geographic factual bias, multilingual and cultural extensions, mitigation attempts, and

LLM use in policy and health decision contexts.

2.1. Geographic factual bias and multilingual extensions

[Manvi et al. \(2024\)](#) established that frontier LLMs carry systematic biases against regions of lower socioeconomic conditions, with correlations up to 0.70 between model-generated ratings and a proxy for those conditions; the outcome space is subjective ratings, which leaves open whether the same bias appears in objective recall of regulatory facts. [Moayeri et al. \(2024\)](#) quantified factual recall against World Bank indicators across 20 LLMs, documenting $1.5\times$ higher absolute relative error for Sub-Saharan Africa than for North America; the design is bounded by the macro-economic indicator space, while an environmental manager asks narrower, source-specific questions whose ground truth lives in environmental councils and monitoring agencies. [Mirza et al. \(2024\)](#) showed that newer GPT-4 versions do not monotonically improve and that the privileging of Global North statements persists.

The multilingual probing literature established early that what a model recalls depends on the language of the query. [Jiang et al. \(2020\)](#) probed cloze-style facts in 23 languages and found accuracy below 15% even in English and Spanish and below 5% in Marathi and Yoruba, with half of the correctly recalled facts recalled in one language only; [Kassner et al. \(2021\)](#) translated the same benchmarks into 53 languages and found mBERT below 60% of its English performance in 32 of them, a bias toward entities of the query language, and no clear relation between the size of a language’s Wikipedia and its recall. Those results concern masked encoders of 2020; the question they raise, whether recall in a language tracks the corpus of that language, is the one H2 and H4 put to generative models. [Myung et al. \(2024\)](#) covered

16 countries and 13 languages in everyday cultural knowledge, finding that LLMs perform better in the native languages of high-resource cultures and worse in those of low-resource ones, the finding that motivates H2 and connects to the resource hierarchy of [Joshi et al. \(2020\)](#). Regional benchmarks have proliferated: BRoverbs ([Almeida et al., 2025b](#)), TiEBE ([Almeida et al., 2025c](#)), ALBA ([Vieira et al., 2026](#)) and AMALIA ([Simplício et al., 2026](#)) document concrete failures in culturally grounded tasks but do not address applied policy use. Work on Portuguese corpus construction is explicit that the effort has centred on English: [Almeida et al. \(2025a\)](#) assemble a 120B-token Portuguese corpus because language-specific pipelines are needed to reach industrial quality. That asymmetry in how much text exists per language is what H4 tests.

These studies establish that the phenomenon exists and is measurable. None isolates a single applied domain with verifiable official ground truth, and none treats prompt framing as a manipulable factor.

2.2. Mitigation, persona, and role conditioning

[Opuszek and Böhm \(2026\)](#) found that reasoning helps in aggregate but does not eliminate regional disparities, and [Kerche et al. \(2026\)](#) proposed a place-based typology situating geographic bias in a broader sociotechnical landscape. Both are conceptual and aggregate.

Role conditioning is what the vendors instruct their own users to do. Anthropic documents role setting in the system prompt ([Anthropic, 2026](#)), OpenAI’s guide opens the recommended message structure with an identity block ([OpenAI, 2026](#)), Google’s Gemini documentation asks that role definitions, which it calls persona, be placed in the system instruction ([Google,](#)

2026), xAI’s standard example initialises every conversation with a role-assigning system prompt (xAI, 2026), and the pattern is catalogued in the prompting literature (White et al., 2023). The systematic evidence on whether it helps is negative: across four LLM families and 2,410 factual questions, expert personas did not improve accuracy over a no-persona control and slightly reduced it across 162 roles (Zheng et al., 2024). An independent study with six models on graduate-level multiple-choice questions reaches the same conclusion: matching an expert persona to the problem type had no significant effect on performance (Basil et al., 2025). Whether persona conditioning helps or harms factual reliability in high-stakes domains is therefore unsettled and, to our knowledge, untested for geographic equity. We treat persona as a pre-specified experimental factor and test both the Norm-Compliance hypothesis (persona reduces the gap) and the Persona-Vacuum hypothesis (persona amplifies it).

2.3. LLMs in policy, health, and the administration

Le Coz et al. (2025) evaluated LLM policy recommendations for homelessness across four cities (Barcelona, Johannesburg, South Bend and Macau) and read the pattern as “a context-blind rigidity and a potential bias based on the published online discourse on homelessness in the regions of the world with overrepresented data in LLMs”, against experts who tailored decisions to locally encoded realities. That is one domain and one task type, which we generalise to five task types spanning recall, synthesis and recommendation. The closest precedent in a regulated domain is Dahl et al. (2024), who asked four public models verifiable questions about United States federal cases and found hallucination rates between 58% and 88%, highest for the lowest courts

and least prominent jurisdictions, and highest again for the oldest and the most recent cases, so that the models’ knowledge lags the doctrine in force; our recall floor and the ladder analysis of Section 4.9 are the environmental counterpart of those two gradients. In health, [Kozlakidis et al. \(2026\)](#) argue that hallucinations carry acute risk in regulatory environments with limited technical capacity, where language and cultural mismatch increases error, a risk that meets the mortality burden documented by the Global Burden of Disease estimates ([GBD 2019 Risk Factors Collaborators, 2020](#)).

The deployment question is no longer hypothetical. [Kim \(2026\)](#) estimates bureaucratic productivity effects of generative AI quasi-experimentally, and [Robinson \(2026\)](#) documents that choosing which model an agency runs is now an explicit procurement decision between open-weight and vendor-hosted options. Our model-level findings speak to that decision: open-weight models as a group trail the closed ones by +12.8 points on the composite even though one open-weight model (DeepSeek-V3) leads the roster, and the regionally tuned Portuguese model is the weakest of the fourteen. Two further strands frame why a wrong answer matters administratively. [Choroszewicz \(2026\)](#) shows that generative-AI support tools position frontline practitioners as “moral crumple zones”, absorbing responsibility for failures originating upstream, and [Cordella and Gualdi \(2024\)](#), analysing the first binding measure a regulator imposed on a generative-AI service (Italy’s intervention on Chat-GPT), argue that technology-neutral frameworks overlook what is specific to these systems and that regulators must engage their technical workings directly. On reception, [Aoki et al. \(2024\)](#) test whether the type of explanation changes perceived accuracy, fairness and trustworthiness among those

affected.

None of this establishes whether the underlying answer is correct, or whether correctness is distributed evenly across the states that rely on it. Perceived trustworthiness and measured accuracy are different quantities, and a system can be well explained, well governed, well received, and wrong. That is the gap this study addresses, in a domain where the correct answer is published in an official register and can be checked.

2.4. Positioning of the present work

The design differs from prior work in four ways: a single high-stakes applied domain with verifiable official ground truth rather than numeric trivia or cultural knowledge; persona as a pre-specified within-subject factor rather than an uncontrolled framing choice; theory-driven country sampling on independent axes (the UNCTAD developed/developing split, Joshi linguistic resource, development) rather than regional convenience; and an explicit corpus-representation mechanism test with sensitivity analysis. No prior study combines a regulated public-health domain, a persona manipulation, and a pre-specified mechanism test.

3. Methods

3.1. Research design

The study is a pre-specified, factorial, multi-country benchmark experiment quantifying geographic and persona-modulated bias in 14 large language models on air pollution policy tasks. The pre-specified design crosses 15 countries (stratified on three theoretically grounded axes), 14 LLMs spanning five access tiers, one domain, five task types and two persona conditions,

with two stochastic replications per cell; a post-hoc extension (Section 3.3) enlarges the sample to 25 countries. The stimulus set crosses each country’s five tasks and two personas in English, plus a within-country native-language rendering for the nine countries with a target native language, scored on the $[0, 1]$ composite. The protocol fixed, before collection, the hypotheses, primary models, persona protocol, exclusion criteria and multiple-comparison corrections. A Brazil-only collection executed under an earlier three-domain design is reclassified as an extended pilot and reported separately; it informed prompt calibration and contributes no observations to the main analysis.

3.2. Country selection

Countries were selected by theoretical sampling (Patton, 2015) triangulated across three independent classifications: the Global North/South distinction, operationalised through UNCTAD’s developed/developing economies classification (UNCTAD, 2024), which carries the coloniality-of-knowledge framework motivating the study (Mohamed et al., 2020); the Joshi et al. six-class taxonomy of linguistic-resource representation (Joshi et al., 2020); and World Bank income groups (World Bank Group, 2025).

The pre-specified sample of 15 (Brazil, Mexico, Argentina, Peru; Nigeria, South Africa, Kenya, Egypt; India, Indonesia, Bangladesh, the Philippines; and the United States, Germany, Japan as Global North references) covers four world regions and four income groups. The linguistic-resource axis turned out far less differentiated than intended: the dominant official languages fall into only three Joshi classes, twelve of the fifteen sharing Class 5, and extending to 25 does not help, since eighteen are Class 5. This is a property of the world rather than of our draw, because the official languages of

most states are themselves high-resource, but it leaves the linguistic axis too little variance to be separated from the others, and we treat the Joshi results accordingly (Section 4.6). Countries with compromised statistical governance or without a publicly retrievable national standard were excluded from the pre-specified fifteen; Oceania entered only with the extension (Australia), and Joshi Class 0 languages are not represented, which we note as a scope limitation.

3.3. Sample expansion after the pre-specified analysis

After the pre-specified 15-country analysis, and reported here as a post-hoc extension rather than as part of the locked protocol, the sample was enlarged to 25 countries under identical procedures, to raise power for the gradient and mechanism tests and to multiply the within-country language pairs available for H2. The manuscript reports the 25-country sample throughout; the primary family recomputed on the pre-specified 15 alone is tabulated in Supplementary Material, so that a reader can see what the locked protocol would have concluded. Because the extension followed observation of the initial results, no pre-specified conclusion is revised on the extension alone, and the expansion is treated as a power-and-robustness check rather than as a new test. Angola, like Nigeria and Argentina, has no national ambient PM_{2.5} standard, adding no-standard cases in which a faithful model must decline to state a non-existent threshold. The ten additions and their roles are detailed in Supplementary Material. Two extra question variants per (country, task) were generated for the H2 pairs at design time but were never collected or scored; the H2 analysis uses the single native-language rendering of each English prompt.

3.4. *Country covariates*

Human Development Index (UNDP 2023–24) and GDP per capita index developmental position and enter the mechanism analysis as adjustment covariates. The corpus-representation exposures divide along a distinction the mechanism analysis makes explicit, because the two channels apply to different effects. Language-corpus measures quantify how much pretraining text exists in a language and are the candidate mechanism for the native-language penalty: mC4 token count (Xue et al., 2021, Table 6), OSCAR-2301 corpus size (corpus introduced by Abadji et al., 2022; per-language figures read from the OSCAR-2301 dataset card, whose totals differ from the 22.01 release described in that paper), and the Joshi class (Joshi et al., 2020). Country-corpus measures quantify how much the encyclopedic corpus is about a country, independent of language, and are the candidate mechanism for the geographic gap: the English Wikipedia article length, the number of Wikipedia language editions with an article on the country, and the number of Wikidata statements on the country entity.

The two families are not interchangeable. Because the geographic gap is measured in English, with language held constant across countries, a residual North/South gap cannot be a language-corpus effect. Wikipedia and Wikidata are never ground truth here; they are the explanatory variable. What an answer is scored against is always an official register (Section 3.9). Had we scored against an encyclopedia, the study would measure agreement with an encyclopedia, which is not the question a public administration faces.

3.5. Model selection

Fourteen LLMs were benchmarked across five access tiers, spanning roughly two orders of magnitude in parameter count and four training-data geographies (United States, China, Europe, Brazil): open-weight frontier (5), open-weight mid (3) and open-weight small or regional (2), closed-accessible (3) and closed-frontier (1). Three of the open-weight models were served locally (Ollama, pinned tags: Phi-4 14B, Qwen3 14B, Cabra-Mistral 7B); the other seven open-weight models were served through hosted endpoints (Groq free tier for five, the DeepSeek and Cohere APIs for two), and the four closed models through their vendors’ APIs. The small tier includes a Brazilian-Portuguese instruction-tuned model (Cabra-Mistral 7B v3, a fine-tune of Mistral-7B-Instruct-v0.3), tested against a scale-matched globally trained open model for H3. Each model was queried with an identical pinned tag throughout the collection window. The full roster is in Supplementary Table S1.

We compare models as served through a fixed access stack, not idealised weights. Provider infrastructure, serving configuration and API layer are part of what each label denotes, and a residual gap could in principle reflect the stack rather than the weights; we hold this constant by pinning version tags and a fixed sampling configuration per provider throughout a fixed window (Section 3.10).

3.6. Domain and task taxonomy

The single domain is air pollution policy, chosen because it anchors to officially published, verifiable ground truth and maps to concrete decision

activities of a public environmental manager. Each of the five tasks corresponds to a question a manager has to answer in the course of ordinary work, and each was drafted so that a wrong answer would carry an administrative consequence. Table 1 gives them with the prompt exactly as issued to the models, using Brazil as the worked instance, alongside the decision each stands in for.

The tasks are ordered by how much of the answer is fixed by an official source. T1, T2 and T4 admit verifiable ground truth and carry the primary accuracy signal; T3 and T5 are rubric-scored open-generation tasks, where competent answers legitimately differ. The ordering lets us ask whether models fail uniformly or specifically where a published national value has to be recalled, which is the question the results turn on. The country-by-task mapping of official source classes and verifiability status is in the ground-truth registry (Section 3.9), and the full 25-country prompt set is released with the data.

3.7. *Persona manipulation*

Persona is a within-prompt factor with two levels, and the manipulation follows the practice the vendors themselves document and recommend (Section 2). The neutral condition presents the query as a plain information request; the `public_manager_env` condition prepends a fixed role frame identifying the requester as a municipal or regional environmental management official, holding the substantive question identical. For Brazil’s T1 the frame is the single sentence *“I am the municipal environmental management official of São Paulo, Brazil, and I need to brief decision-makers.”* A single template is applied uniformly across countries and tasks, translated

where the prompt language is non-English, so that any between-condition difference is attributable to the frame rather than to question wording. Persona order within each prompt-model cell is randomised, and a manipulation check verifies that the frame is acknowledged in the response. Exact wording, translation procedure and manipulation-check coding are in Supplementary Material.

One implementation detail matters for how the result should be read. Every request in this study is sent as a single user turn with no system prompt, so the role frame is prefixed to the user message rather than installed as a system instruction. Three of the four vendor guides place the role in the system prompt; Google’s states that it belongs either in the system instruction or “at the very beginning of the user prompt” ([Google, 2026](#)), which is the channel we test. The choice is deliberate: the system-prompt route requires API access and a developer to configure it, while the public servants whose decisions motivate this study reach these models through the consumer chat interface, where the only available channel is the message they type. We therefore test role framing as the sector performs it, and [Section 5.7](#) states what remains untested. Persona is the basis of H6.

3.8. Prompt construction and validation

Prompts were authored for the domain by the research team with domain supervision and validated against the ground-truth registry before entering the main stimulus set, with every factual claim in a registry entry required to resolve to an official source URL or a peer-reviewed DOI before the corresponding prompt was admitted. The analysis plan called for a 50-prompt pilot scored by two independent human raters ($\alpha \geq 0.70$) before scaling the

design. That pilot was not run, and no rater-level α exists for this study; rubric reliability is evidenced instead on the main data, by a three-vendor panel over every item that required a judge (Section 4.11), which returns $\alpha = 0.527$. We report this as a departure from the plan rather than as a threshold met.

All prompts were authored in English and, for a stratified subset of countries, additionally in a relevant native language (Supplementary Section S3). Native-language renderings were produced by an LLM translator (Claude Sonnet 4.6) prompted to preserve meaning, persona frame, proper nouns and acronyms, with the factual ground truth held in English since the target value is language-invariant. We did not employ human native-speaker translators, which remains a limitation (Section 5.7); we audited the renderings after the fact by back-translating all 90 native prompts with a different model and having two further models compare each against the English original (Section 4.4). The native language is the language of the user, not necessarily of the instrument: Brazil’s, Portugal’s and the five Spanish-speaking countries’ standards are published in the prompt language, whereas India’s National Ambient Air Quality Standards are published in English, so the Hindi condition asks in a language the official text does not use.

3.9. Ground truth and the ground-truth registry

Ground truth is a set of four task-specific registries, released with the data, and we describe each as it was built rather than as it was planned (Supplementary Table S14 records the deviations). For T1 the key is the annual $\text{PM}_{2.5}$ value in force during the collection window, taken from the national instrument itself (statute, decree, regulation or gazette) and stored

with the source, a verbatim excerpt and a status flag. Twenty-one countries have a numeric key. Three (Angola, Argentina, Nigeria) have no national $\text{PM}_{2.5}$ standard, so the correct answer is to say so, and a stated value is scored as fabricated. Egypt has no key, because the official Arabic text could not be retrieved, and is excluded from T1 (56 cells). Bangladesh is scored against the $35 \mu\text{g}/\text{m}^3$ of its Air Pollution (Control) Rules 2022, which replaced the 15 of the 2005 gazette; an earlier version of the registry carried the 2005 value, and the correction is logged with the released files. Where an instrument sets a ladder of staged values (Brazil’s five CONAMA stages, the EU limit of 25 with 10 from 2030, the United Kingdom’s 20 with an English target of 10 by 2040, the United States’ 9.0 succeeding 12.0, Bangladesh’s 35 succeeding 15), the registry stores the whole ladder as well as the value in force, so that citing a neighbouring stage can be separated from fabrication (Section 4.9). Canada’s CAAQS and the Philippine value are objectives rather than legally binding limits; they are the only national reference values those countries publish and are treated as the key, which the text notes where it matters. The registry does not record the date on which each instrument entered into force, so the distance between that date and a model’s training cut-off could not be entered as a covariate; the ladder analysis in Section 4.9 is the nearest available substitute.

For T2 the key is the most recent annual mean $\text{PM}_{2.5}$ concentration that the WHO Ambient Air Quality Database (v6.1, 2024) records for the prompted city, which compiles the values reported by national and subnational agencies; any reported year is accepted within $\pm 20\%$, so that a model citing an agency value from a different year is not penalised for the year.

The window is not tier-neutral by construction, because Global North cities have longer WHO series (7.8 accepted years on average against 5.6), but the range of values it admits is similar in the two tiers (ratio of the largest to the smallest accepted value 1.46 against 1.43). Two cities have no PM_{2.5} entry (Luanda, Cairo) and those cells fall to the judge. For T3 the key is the WHO Global Health Observatory estimate of deaths attributable to ambient air pollution (indicator AIR_41, 2021) with its uncertainty interval widened by half on each side as the accepted range, plus the published ordering of causes; the indicator covers ambient air pollution as a whole while the prompt asks about PM_{2.5}, and a model citing a different burden study (the GBD, a national cohort) is scored by the judge on whether its figure and sources are defensible, not against the WHO number. For T4 the reference set is the UNEP Global Air Pollution Legislation catalogue for each country, with manual overrides where it is stale. The registry records source, URL, access date and status per entry, but it has no human-validator field and no per-document hash: verification was documentary, by the authors, against the primary text. Entries without an equivalent official source are excluded from the code-adjudicated comparison for that cell, so that missing ground truth is never scored as a model error.

3.10. Data collection

Calls were executed at temperature 0.3, with two replications per prompt-model-persona combination, inside a fixed window to minimise model-update confounds; the reported accuracies are therefore point-in-time estimates. Two exceptions are declared. The GPT-5 family rejects any temperature other than the default, so GPT-5 and GPT-5-mini ran at temperature 1.0

with reasoning effort set to minimal; and no seed was sent to any provider, an earlier draft having stated otherwise, so the two replicates at temperature 0.3 bound, rather than remove, within-model variance (Supplementary Material compares the headline effects with that variance). Every response was stored with complete request metadata (pinned tag, timestamp, token counts, finish reason, persona condition) under SHA-256 checksums. No response was excluded for its length or its language: empty responses are scored as returning no value, and a response in a language other than the one requested is scored on its content. Empty responses are more frequent under native-language prompts (4.2% of English responses on the three code-scored tasks against 11.4% in Spanish, 13.9% in Portuguese and 11.5% in Hindi), which is part of the language effect rather than attrition from it.

Per-cell coverage is not uniform: free and rate-limited endpoints returned fewer admissible responses than metered APIs, so the number of scored English-prompt cells ranges from 261 to 500 per model (777 for the model with the most native-language pairs when those are counted). We report each model’s realised N alongside its score (Table 2) rather than dropping under-covered cells, the tier gap is re-estimated on the prompts every model answered (Section 4.6), and the full coverage matrix, with quota and exclusion reasons for every shortfall, is in Supplementary Material.

3.11. Measures

Scoring assigns each task the most reliable instrument it admits.

Where the answer is a number in an official register, code decides. For T1, T2 and T3 the returned value is extracted and compared with the registry by a deterministic scorer, with plausibility guards that reject implausible

extractions (a four-digit year read as a death count, a national population read as a mortality figure, a 24-hour value read as an annual one). The extractor normalises Devanagari digits, decimal commas and units written out in words, and its behaviour on the cases that matter for the language comparison (a comma decimal, a Hindi numeral, a parenthetical daily value beside the annual one) is fixed by a unit-test suite released with the code; a stratified manual audit of native-language T1 responses it classified as carrying no value is reported in Supplementary Material, and no equivalent audit was run for T2 and T3. Comparing a value with a key is exact, whereas a language model asked to do the same arithmetic reproduces it with noise. Code resolves 99.8% of the T1 cells in the analysis (1,614 of 1,617), 56.3% of T2 and 71.8% of T3; a T1 response with no extractable value counts as not having returned it. For the three no-standard countries, code scores whether the answer states that no national standard exists (correct), asserts a value (fabricated) or declines to answer (neither), and the tier comparison is reported with and without them.

The code verdict replaces one subcomponent of the composite, factual accuracy (weight 0.30); the other four subcomponents of a T1, T2 or T3 cell remain those assigned by the judge that scored it. For T1 that judge is the collection judge, so the composite on T1 still carries the judge-and-tier confound described below in 0.70 of its weight. For that reason the primary outcome for the tier comparison on the register-valued tasks is the binary code verdict itself, whether the answer returned the value in force, and the composite is reported as the secondary outcome (Section 4.6).

Where judgement is required, a three-vendor panel decides by its mean.

The residual of T2 and T3, with all of T4, is scored by Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3, chosen for vendor diversity so that intra-vendor correlation does not inflate apparent agreement. T5, the open recommendation, retains the scores of the original collection judge; that judge was GPT-5-mini for the fifteen pre-specified countries and Claude Haiku 4.5 for the ten added in the extension, and because the extension holds seven of the ten Global North countries, judge and tier are confounded in any effect that still rests on those scores. This is why the factual verdict on T1 was moved to code: on the original scores the Global South deficit in T1 was 0.21 against 0.38 under GPT-5-mini but 0.37 against 0.44 under Claude Haiku, a difference between judges as large as the difference between tiers (Supplementary Material). T5 carries no tier effect in either instrument. Reliability is reported as Krippendorff’s α , the single-judge ICC(2, 1) and, as the operative quantity, the ICC(2, 3) of the panel mean, a two-way random-effects, absolute-agreement coefficient in which the judges’ different severities count against it, computed on every item the panel scored rather than on a calibration subset. Language-model judges agree with human raters at roughly the level humans agree with each other on open-ended tasks (Zheng et al., 2023), but they carry position, verbosity and self-enhancement biases, and models recognise their own outputs well enough to prefer them (Zheng et al., 2023; Panickssery et al., 2024). One panel member (DeepSeek-V3) is also an evaluated model, and we test for self-favouring directly; the other two judges belong to vendors whose models are in the roster (Anthropic, Google), so a vendor-level preference cannot be excluded by that test. No human gold-standard layer was collected; ground truth is anchored to official

primary sources (Section 3.9).

The unit of analysis is the scored cell, not the log line. Duplication exists at two levels and each has a stated rule. At the response level, 21.9% of (model, prompt, replicate) keys have more than one stored answer, because collection runs were resumed and re-run; the answer that enters the analysis is the first stored in run-file order, which is the one the judge scored, and the code verdicts are computed on that same answer so that the two instruments describe one response. Scoring every stored answer instead and averaging the verdicts changes the code verdict in 3.1% of cells and the mean by 0.0002 (Supplementary Material). At the score level, 951 cells (11.5%) carry more than one judge score, because the judge was re-run on parts of the set on different dates; those are repeated measurements of the same item and are collapsed by their mean, leaving 8,300 cells, or 8,244 once the 56 Egyptian T1 cells without a key are removed (6,573 English-prompt). Treating re-runs as independent would inflate n and give extra weight to whichever cells happened to be reprocessed. The rule was fixed on methodological grounds and does not depend on the result.

The primary composite combines five pre-specified subcomponents (Supplementary Section S6), each normalised to $[0, 1]$: factual accuracy against the registry entry (weight 0.30), contextual completeness (0.25), citation quality (0.15), calibration (0.15) and hallucination absence (0.15). Every headline effect is recomputed under equal weights and a PCA-derived weighting, and a conclusion counts as robust only if its direction is stable across all three. The composite is computed per (country, model, persona) cell to support the H6 difference-in-differences test.

3.12. Analytical strategy

The three primary tests are pre-specified as a single family (F1) and corrected together with Bonferroni-Holm. Secondary tests (F2) are reported unadjusted within family, and declared exploratory tests (F3) use Benjamini-Hochberg FDR at $q = 0.10$ (Benjamini and Hochberg, 1995), whose guarantee holds under positive dependence; the Benjamini-Yekutieli correction, valid under arbitrary dependence, returns the same decisions on this family, since the F3 tests it covers have p -values several orders of magnitude below the threshold.

H1 (geographic gradient, primary) regresses country-level mean composite accuracy on the Joshi/HDI gradient via Spearman ρ , pre-specified one-sided at $\rho \geq 0.55$, with a Mann-Kendall trend test as a mandatory non-monotonicity complement. The binary Global North/South contrast on the same country means was pre-specified as a sensitivity analysis of H1, and its decomposition by task (Section 4.6) was not pre-specified and is exploratory. H4 (corpus mechanism, primary) relates corpus representation to accuracy by partial Spearman ρ adjusting for HDI and log GDP per capita, with Wikipedia counts as the pre-specified proxy and Wikidata coverage as exploratory complements. H6 (persona, primary) is a country-level difference-in-differences, $\Delta_{\text{GS}} - \Delta_{\text{GN}}$, supported if the persona condition reduces the Global South gap by at least 0.05 on the composite, with inference by a 5,000-permutation country-level test; because the pre-specified test cannot distinguish a null from an inconclusive result, we add an equivalence test, two one-sided tests on the country-level bootstrap distribution of the difference-in-differences with margins of ± 2 and ± 5 pp. H3 (secondary) is a one-sided

Mann-Whitney contrast (Cliff’s δ). H2 and H5 (exploratory) are reported descriptively under F3 correction.

Wherever a test in the Results is first run on responses, we also report it with the country or the model as the unit, because responses are nested in 25 countries and 14 models and a response-level p -value overstates the evidence for any contrast that varies between countries. Three cross-cutting analyses corroborate the tier gap: a Gaussian linear mixed model (`statsmodels` MixedLM, REML) with crossed random intercepts for country and model, adjusting for centred HDI; a Bayesian hierarchical re-estimation (`pymc` (Abril-Pla et al., 2023), weakly informative priors); and E-values for robustness to unmeasured confounding, computed for the standardised mean difference in composite accuracy between tiers through the approximation $RR \approx \exp(0.91 d)$ that VanderWeele and Ding (2017) prescribe for standardised effects and whose assumptions it inherits. The corpus mechanism is probed with an exploratory mediation model (`semopy` (Igolkina and Meshcheryakov, 2020)), and the persona manipulation with a role-frame-acknowledgement check. Pre-specified robustness analyses appear in Section 4.12.

3.13. Study protocol and ethics

The analytic dataset, after quality gates but before model fitting, is frozen and hashed, so that the analysis can be audited against the locked specification rather than taken on trust. Release terms are stated at the end of the paper.

No personal data from end-users of LLMs were collected; the study analyses model outputs to publicly relevant policy questions. Provider terms of

service were reviewed and respected, and responses are redistributed solely for research under fair-use and reproducibility provisions, excluded from any commercial application.

4. Results

Sample note. Results are reported on the 25-country sample. The pre-specified benchmark covered 15 countries and was extended post hoc by ten (six Global North references, the United Kingdom, Canada, Australia, South Korea, France and Italy, and four native-pair countries, Colombia, Chile, Portugal and Angola). Fourteen LLMs are scored across five tasks, two personas, and four languages (English plus a country-native language, Spanish, Portuguese or Hindi, for the nine applicable countries, giving the within-country English/native pairs that test H2). The primary family recomputed on the pre-specified 15 countries alone is tabulated in Supplementary Material, and no pre-specified conclusion is revised on the extension alone; where the two samples differ on a hypothesis the paper turns on, the text says so. Scoring is adjudicated by code wherever the answer is a number checkable against an official register, and by the mean of a three-vendor judge panel elsewhere (Section 4.11); the T1 ground truth is anchored to the national instrument in force for 24 countries and has no key for Egypt, which is excluded from T1 (Section 4.9). Every test first run on responses is also reported with the country or the model as the unit.

4.1. Sample and scoring

The analysed corpus comprises 8,244 scored cells on the $[0, 1]$ composite, of which 6,573 are English-prompt responses across the 25 countries; the

remainder are the within-country native-language pairs used for H2. Code adjudicates the factual verdict against the official register wherever the answer is a published number, resolving 99.8% of T1, 56.3% of T2 and 71.8% of T3 cells; the residual, with all of T4, is scored by the mean of a three-vendor panel, and T5 retains the scores of the collection judge. In total 6,434 cells carry a code or panel score.

The panel mean replaces the single-judge design of the original protocol, in which the collection judge (GPT-5-mini for the pre-specified fifteen countries, Claude Haiku 4.5 for the ten extension countries) was itself among the evaluated models and, because the extension holds seven of the ten Global North countries, was confounded with tier. Judges differ in severity (mean composite 0.617 for Gemini, 0.451 for Claude, 0.379 for DeepSeek), so a single judge fixes an arbitrary level, and single-judge reliability is $\text{ICC}(2, 1) = 0.56$ against $\text{ICC}(2, 3) = 0.79$ for the panel mean. One panel member (DeepSeek-V3) is also an evaluated model, so we checked for self-favouring: it scores its own outputs more harshly relative to the other two judges (-0.177) than it scores any other model’s (-0.123 to -0.160).

4.2. Accuracy by model

Table 2 orders the 14 models by mean composite: frontier systems lead and small open-weight models trail. On H3, the Brazilian-Portuguese regional model Cabra-Mistral 7B is the weakest of all (0.351 against 0.557 for the rest; Mann-Whitney $p \approx 0$ on cells, Cliff’s $\delta = -0.48$). Cliff’s δ is the probability that a randomly drawn answer from one group beats one from the other, rescaled to $[-1, +1]$, so -0.48 means the regional model loses roughly three such comparisons for every one it wins. With the country as

the unit, Cabra trails the mean of the other thirteen in 25 of 25 countries by -20.6 pp on average (Wilcoxon $p = 1 \times 10^{-5}$). The pre-specified test was narrower, Cabra against its scale-matched peer Llama 3.1 8B on Portuguese-language prompts, and it points the same way with a smaller margin: on Brazil the regional model scores 0.258 against 0.313 ($\delta = -0.26$, $n = 20$ cells each), and across the three Portuguese-speaking countries 0.304 against 0.362 ($\delta = -0.17$, $n = 60$ each); paired by prompt with the replicates averaged, Cabra loses on 20 of the 30 Portuguese prompts, by -5.9 pp (Wilcoxon $p = 0.002$). Two readings of this have to be separated, and only one of them is a finding. That a 7B model trails frontier systems an order of magnitude larger is a statement about scale. The pre-specified question was narrower and is answerable: does tuning a base model on Brazilian Portuguese improve what it recalls about Brazil? Re-served on the same 30 Portuguese items, Cabra and the base model it was fine-tuned from return the register value at 47% and 50% (Section 4.12). Regional instruction tuning at this scale does not improve register-value recall over its own base, and the composite margin against Llama 3.1 8B comes from the judged tasks and subcomponents rather than from the values the code checks.

4.3. Accuracy by task: the recall floor

Difficulty is task-dependent (Table 3). The hardest task is T1 (technical-standard recall, 0.393), which requires returning the national annual PM_{2.5} standard in force. The open recommendation task (T5) scores highest (0.769) because its rubric rewards plausible, well-structured policy advice that does not hinge on a single fact. Pooling the two recall tasks (T1, T2; mean 0.434) against synthesis and recommendation (T3–T5, 0.612), the difference is large

on cells (Mann-Whitney $p \approx 0$, Cliff’s $\delta = -0.41$, roughly two and a half losses for every win) and holds with the model as the unit (recall below the rest in 14 of 14 models, by -17.6 pp, Wilcoxon $p = 1 \times 10^{-3}$) and with the country as the unit (25 of 25, -18.1 pp, $p = 1 \times 10^{-5}$). Models are weakest where applied environmental management needs them most, in recalling the regulatory threshold in force.

The two corrections of the scoring are informative here. T2 was originally scored against a placeholder key and rises from 0.368 to 0.473 once each returned concentration is compared with the authorised range by code. T1 always had an official key but was scored by two collection judges of different severity, and moving its factual verdict to code raises it from 0.367 to 0.393 while keeping it the hardest task; the code verdict also shows what the judges could not, that more than a third of the wrong T1 answers (281 of 759) cite a value that sits on the country’s official ladder, a superseded value or a future stage, rather than a fabricated one (Section 4.9). The floor is shallower than first reported and concentrated in normative recall.

4.4. *H2: asking in the local language lowers accuracy*

For the nine countries with a target native language, each prompt is rendered in both English and the native language with content held fixed, so that the contrast is within-country and within-model. Replicates are averaged within each (model, prompt) cell before pairing, since pairing replicate to replicate would inflate n . Across the 839 matched cells, native-language prompting lowers mean composite accuracy by 5.0 percentage points (Wilcoxon signed-rank $p = 5 \times 10^{-16}$). That test treats cells as independent, and they are nested in nine countries and fourteen models, so

we also report the inference that respects the nesting: a mixed model on the paired differences with a random intercept per country gives -5.0 pp (SE 1.0, $p = 6 \times 10^{-7}$), with a random intercept per model -4.8 pp ($p = 2 \times 10^{-9}$); with the country as the unit ($n = 9$) the penalty is negative in 8 of nine and averages -5.0 ± 1.1 pp ($t = -4.6$, $p = 0.002$); with the model as the unit ($n = 14$) it is negative in 12 of fourteen (Wilcoxon $p = 0.058$). All three languages carry a significant penalty (Table 4): Hindi -11.4 pp, Spanish -4.5 pp, Portuguese -4.2 pp. Hindi is the language with the smallest corpus and carries the largest penalty; Spanish and Portuguese, whose corpora differ in size, carry penalties that do not differ. India, which scores highest of all 25 countries in English, drops furthest in Hindi, and part of that drop may be regression to the mean, since a country whose English score is highest partly by chance has further to fall; with a single Hindi country the two readings cannot be separated.

H2 carries the study’s central claim, and it survives every attempt to remove it: it is not driven by any one country (leave-one-out -4.3 to -5.5 pp, all significant), by any one model (-4.5 to -5.2 pp), or by outlying cells (5% trimmed -4.5 pp, $p = 6 \times 10^{-20}$); it holds under every weighting of the composite; and it appears in every task and under both persona conditions. Nor is it silence. Native-language prompts return an empty response more often than English ones, so the penalty could in principle be a count of answers that never came; dropping every cell whose stored response was empty leaves it at -5.15 pp ($n = 819$ pairs, $p = 4 \times 10^{-17}$), marginally larger rather than smaller. The penalty is what the model says, not what it fails to say.

Two further checks answer the objections a sceptical reader raises first. The first is that language-model judges might penalise non-English prose for style rather than content, so we test a binary outcome no judge touches: does the response contain a value the extractor can compare with the register? The extractor reads Devanagari numerals, decimal commas and units written out in words, so a Hindi or a Spanish-language number is not penalised for its notation (Section 3). English prompts yield a verifiable value 71.9% of the time and native-language prompts 62.5%, a paired difference of -9.4 pp, with the native version worse in 134 of the 218 discordant pairs (sign test $p = 9 \times 10^{-4}$): Hindi -39.5 pp, Spanish -13.4 pp, while Portuguese, the best-resourced of the three, shows no such loss ($+3.5$ pp). Portuguese responses return a number as readily as English ones; the number is more often wrong. On the code verdict itself (native minus English, cells adjudicated by code in both languages) the Portuguese penalty is -9.1 pp ($n = 132$, $p = 0.019$) and the Spanish one -2.1 points ($n = 165$, not significant), -5.0 pp overall ($n = 318$, $p = 0.048$). Where the language is scarce the model is less likely to return a checkable number at all, and where the language is well resourced the number it returns is less often right.

The second is that the native prompts, produced by an LLM translator, might ask something different. We tested this on all 90 unique native prompts, back-translating each with a different model and having two further models audit the result against the English original. On the two items that could confound the effect, fidelity is complete: no prompt changed the country and none changed the requested quantity (90/90 on both). Restricting H2 to the prompts verified as faithful leaves it unchanged (-4.8 pp,

$p = 2 \times 10^{-8}$) against -5.2 pp among the divergent ones, a difference not distinguishable from zero (permutation $p = 0.80$). Separating the pairs by the instrument that scored the native cell, the penalty is -3.8 pp where code decided ($n = 333$, $p = 0.004$), -8.6 pp where the panel decided ($n = 314$, $p = 4 \times 10^{-22}$) and -1.1 pp, not significant, where the collection judge’s score remains ($n = 192$, essentially T5, the open recommendation with no register value). The code-adjudicated subset is the smaller of the two because code acts only where a number exists, so it conditions on the model having produced a value in both languages and selects away the severest failure mode; the panel subset carries that failure mode, which is why judge leniency cannot be assumed to cancel across languages and why the judge-free outcome above is the one that settles the point.

The applied implication runs against intuition. A manager who suspects an English-language model serves their country poorly, and responds by asking in the national language, makes the answer worse, and worse by the largest margin where the language is least represented in training corpora.

4.5. *H6: persona does not narrow the geographic gap*

The persona factor was fully crossed, so H6 is tested as a difference-in-differences. The public-environmental-manager frame leaves the North/South gap unchanged: $+4.8$ pp under the neutral frame against $+5.3$ pp under the manager frame, a DiD of $+0.5$ pp, far below the pre-specified 5 pp threshold; a country-level permutation test (5,000 permutations of the tier label) gives $p = 0.29$. The pre-specified test cannot distinguish absence of an effect from lack of power, so we add the equivalence test: the country-level bootstrap 90% interval of the DiD is $[-0.4, +1.6]$ pp,

Points below the diagonal: the native-language answer scores worse

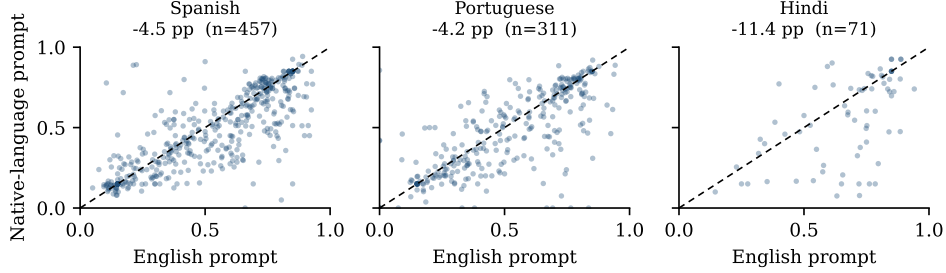


Figure 1: Each point is one (prompt, model) cell scored in English and in the country’s native language, replicates averaged. Points below the diagonal are cells where the native-language answer scored worse.

inside both the ± 5 pp margin of the hypothesis and a stricter ± 2 pp margin, so that the frame is equivalent to no frame within 2 pp on the gap. The persona has a main effect of its own, and it is negative: paired by prompt and model, the manager frame lowers the composite by -0.62 pp ($n = 1,654$ pairs, Wilcoxon $p = 0.003$; with the country as the unit $p = 0.062$). H6 is not supported: presenting the query as coming from a public environmental manager neither narrows nor amplifies the Global South disadvantage, consistent with evidence that expert personas do not improve factual accuracy (Zheng et al., 2024; Basil et al., 2025). A manipulation check finds that 50.2% of persona-condition responses explicitly acknowledge the role frame, so the null is not an artefact of the frame being ignored.

4.6. H1: a tier gap that holds, a gradient that does not

H1 was pre-specified in two parts, a development gradient as the primary test and the binary Global North/South contrast as its sensitivity analysis,

and the two separate cleanly.

The primary outcome for the contrast is the one no judge touches. On the code-adjudicated outcomes (all of T1 and the resolved part of T2 and T3), Global North countries return the register value 0.61 of the time and Global South countries 0.46, a gap of +14.7 pp with a country-level bootstrap interval of [+5.6, +23.9] (permutation of the tier label with the country as the unit, $p = 0.012$). On the composite, the secondary outcome, the ten Global North countries average 0.572 against 0.521 for the fifteen Global South countries, a gap of +5.0 percentage points whose 95% bootstrap interval over countries is [+1.6, +8.4] pp (permutation $p = 0.019$). The composite gap is stable to dropping any single country (leave-one-out [+4.4, +6.0] pp) and to the composite weighting (+4.2 pp under equal weights, +8.9 pp on the factual subcomponent, where it widens). Treating the four European Union members in the sample (Germany, France, Italy, Portugal) as a single unit, so that shared regulation is not counted four times, leaves it at +5.5 pp [+1.6, +9.3] ($p = 0.029$, 7 Northern units against 15). Because seven of the ten Global North countries entered with the extension, we also report the gap where the wave cannot explain it: within the extension alone (seven Global North against Colombia, Chile and Angola) it is +5.9 pp [+2.3, +9.6], and on the pre-specified fifteen (three Global North against twelve) +6.1 pp [+1.8, +10.1], a point estimate of the same size on a sample too small to make it significant by permutation ($p = 0.12$). Coverage is uneven across models, and that unevenness is the gap’s weakest point. Restricting the estimate to the 104 of 248 English prompts that every one of the 14 models answered leaves the point estimate unchanged (+5.1 pp) and the interval, on far fewer

cells, includes zero $([-2.7, +12.8], p = 0.32)$. The composite gap’s significance therefore depends on the unbalanced cells: the effect does not reverse or shrink when coverage is equalised, but this sample cannot demonstrate it on a balanced design. The code-adjudicated contrast above, which pools tasks rather than prompts, is the one that holds with the country as the unit.

Where the gap lives. The decomposition by task was not pre-specified and is exploratory. Re-estimating the contrast on each task separately, with the outcome coded as returning the register value, shows the disadvantage scaling with how much the task depends on a fact specific to that country (Table 5). The table gives two estimates of each odds ratio. The binomial mixed model with crossed random intercepts for country and model returns a conditional odds ratio with a narrow interval; the country-level bootstrap of the raw odds ratio, which treats the country as the unit of inference, returns a wider one and is the honest measure of what 24 countries can establish per task. On the binding national standard (T1), scored by code against the instrument in force, the raw Global South odds of returning the register value are 0.44 of the Global North’s (50.7% against 31.0%), with a country-level interval of $[0.18, 1.05]$ and a country-level permutation $p = 0.062$; the conditional estimate is 0.32 $[0.27, 0.39]$. On policy instruments (T4) the raw ratio is 0.50 $[0.19, 1.15]$ and on health-evidence synthesis (T3), where the answer is an international estimate rather than a national one, it rises to 0.76 $[0.37, 1.46]$. Open recommendation (T5) is at ceiling in both tiers (99.6% against 99.7%), so no contrast is estimable there and the task is reported without an odds ratio. The T1 ratio holds its direction under the two objections its composition invites (Section 4.9): setting aside the three

Global South countries with no national standard the conditional estimate is 0.39 [0.32, 0.49], and on the pre-specified fifteen it is 0.27 [0.21, 0.34]. T4 is the one task in the table scored entirely by the panel, so its ratio could carry a judge’s own ignorance of Southern instruments as well as the model’s; the code-only estimates are the ones free of that.

Two readings of this table have to be kept apart. The point estimates order themselves by how much the answer depends on a fact about the country, which is the pattern the paper’s argument predicts, and the conditional model returns narrow intervals for each of them. But the conditional interval is an interval for a within-country, within-model contrast: the country intercepts alone have a standard deviation of 1.60 on T1 and 2.85 on T4 (Supplementary Material), so the countries are far from exchangeable and a reader who wants a statement about countries needs the country-level interval. Every one of those includes 1 (Figure 2). We therefore report the per-task ordering as a pattern of point estimates and the pooled code-adjudicated contrast as the finding: the gap is real in the aggregate and its concentration on country-specific tasks is exploratory.

At $n = 25$, Spearman $\rho(\text{accuracy}, \text{HDI}) = +0.36$ does not reach conventional significance ($p = 0.076$; Mann-Kendall $p = 0.065$) and stays below the pre-specified $\rho \geq 0.55$ threshold, which is the criterion H1 was fixed against; its Fisher 95% interval, $[-0.04, +0.66]$, includes both zero and the threshold, so the data neither establish the gradient nor exclude one of the size we committed to. At the pre-specified $n = 15$ it is absent ($\rho = +0.08$, $[-0.45, +0.57]$), and no leave-one-country-out re-estimate reaches 0.55 (range $[+0.3, +0.5]$); under equal weights it is 0.35 and under PCA weights 0.38.

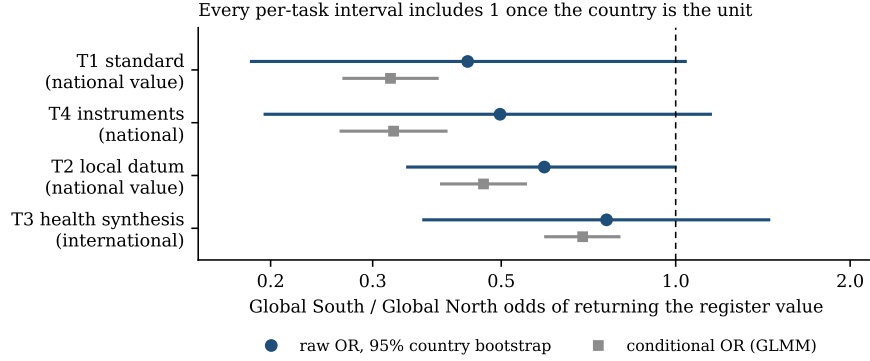


Figure 2: Global South versus Global North odds of returning the register value, by task, under the two levels of inference. Blue: raw odds ratio with its 95% country-level bootstrap interval. Grey: conditional odds ratio from the binomial mixed model. T5 is omitted because both tiers are at ceiling.

We therefore report a tier difference we can demonstrate and a development gradient we cannot: the disadvantage is established as a difference between groups, while its shape across the development range remains open. Under Bonferroni-Holm across the primary family $\{H1\text{-gradient}, H4, H6\}$, no test rejects its null.

The country ordering retains large axis-orthogonal heterogeneity (Figure 3): India (0.651, Global South) tops all countries, Indonesia and South Africa outrank most of the Global North, and Canada and France sit low. The disadvantage is a tier difference layered over substantial country-specific variation, and that heterogeneity is part of why the gradient does not resolve.

4.7. $H4$: corpus representation is not established

The pre-specified mechanism was training-corpus representation, proxied by Wikipedia article and edition count. That proxy is null: Spearman $\rho =$

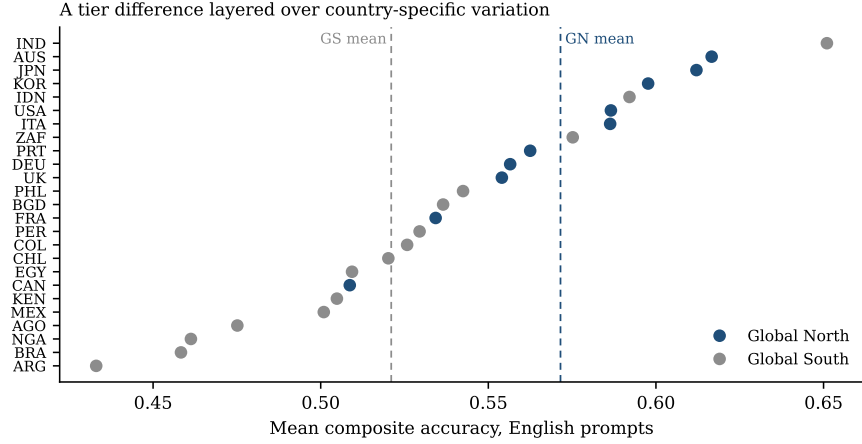


Figure 3: Mean composite accuracy per country on English prompts, ordered. Dashed lines mark the two tier means, whose difference is the composite gap.

+0.05 ($p = 0.82$) uncontrolled, and still null partialling out HDI and log GDP per capita. A post-hoc country-coverage proxy reaches $\rho = +0.37$ ($p = 0.070$) and attenuates after adjusting for HDI (partial $\rho = +0.19$, $p = 0.37$). At country level neither channel is established, because with 25 countries coverage and development are collinear ($\rho = 0.65$ between them).

Changing the unit of variation, and what it can and cannot show

Each country answers five tasks that differ in how much the answer depends on a fact about that country: T1, T2 and T4 do; T3 (an international WHO estimate) and T5 (a generic recommendation) do not. Countries score 0.462 on the first block and 0.658 on the second, a specificity deficit of 0.196. If what limits the model is how much the corpus says about a given country, that deficit should be smaller where coverage is greater, and a fixed effect for country absorbs every country-level main effect, development and income

included, though not their interaction with task dependence. At the response level the interaction between coverage and task dependence is $\beta = +0.031$ (SE 0.005, $p = 6 \times 10^{-10}$, $n = 6,573$), and the pattern across classifications is the one a real effect would produce (Table 6): the purest contrast, T1 against T5, yields the largest coefficient, dropping T5 or T4 leaves it intact, and a placebo inverting the labels returns the exact mirror image.

That response-level p -value is not the evidence, because the interaction varies only between 25 countries, and at the country level the two candidate exposures separate. With standard errors clustered by country, coverage carries $p = 0.099$; development carries $\beta = +0.036$ with $p = 0.027$. The 25 country-level deficits say the same: the deficit correlates with HDI at $\rho = -0.40$ ($p = 0.049$) and with coverage at $\rho = -0.37$ ($p = 0.069$), falling to $\rho = -0.16$ ($p = 0.44$) once HDI is partialled out. Entered together the two are uninformative, because they correlate at $\rho = 0.65$ and neither survives (coverage $\beta = +0.013$, $p = 0.69$; HDI $\beta = +0.027$, $p = 0.35$); within the nine anglophone countries, where the language of the question is the same by construction, the coverage coefficient is $+0.024$ with $p = 0.47$ on nine clusters. The language-corpus proxy entered alongside coverage is null at every level ($\beta = +0.004$, $p = 0.41$), as predicted for responses elicited in English.

What survives with the country as the unit is therefore the development interaction, not the coverage one. The result is narrower than it sounds, and it is not the gradient of H1: a country’s development level does not predict how accurately models answer about it overall (Section 4.6), but it does predict how much worse they answer the questions whose answer is a fact

about that country. Coverage points the same way and does not clear the threshold, and the two cannot be separated in a sample where they correlate at 0.65. We report the pre-specified corpus mechanism as not established, the development interaction as the one country-level mechanism result that clears 0.05, and the choice between the two exposures as open.

The corpus account is visible in H2, where the unit is the language: Hindi, the smallest corpus by mC4 token volume (Xue et al., 2021) and OSCAR size (Abadji et al., 2022), carries the largest penalty, while Spanish and Portuguese carry penalties of the same size despite corpora of different size. With three languages this is descriptive.

4.8. H5: open versus closed-accessible

Closed-accessible models average +12.8 pp above open-weight ones, against the optimistic H5 framing. The open arm includes larger models, so the gap is not attributable to scale alone, although DeepSeek-V3 ranks first overall. H5 is interpreted descriptively.

4.9. Ground-truth provenance, and what the T1 key decides

Every key comes from an official register, anchored by documentary verification and reproducible from the cited URLs: T1 from the national regulation, T2 from WHO Ambient Air Quality Database v6.1, T3 from WHO GHG indicator AIR_41, T4 from the UNEP Global Assessment of Air Pollution Legislation. Three countries have no national PM_{2.5} standard, confirmed from the official text (Nigeria regulates PM₁₀ only, Argentina has no binding federal value, Angola no national legislation), which is where a faithful

model must decline to state a non-existent threshold. Egypt has no retrievable value and therefore no T1 key. All four of these cases are Global South, so the composition of the key is itself a candidate explanation for the T1 gap. It is not: set the three no-standard countries aside and the conditional Global South odds on T1 are 0.39 [0.32, 0.49] of the Global North's, against 0.32 with them. The key also moved under review, in the Southern direction the argument of Section 5 predicts: Bangladesh's entry carried the $15 \mu\text{g}/\text{m}^3$ of its 2005 gazette until an external reader pointed to the 2022 Rules, which set 35, and with the corrected key Bangladesh's T1 accuracy falls from 0.39 to 0.30, because most models cite the superseded value. The full registry is in Supplementary Material. Wikipedia and Wikidata appear only as the explanatory variable of the mechanism test; no answer is scored against them.

The key in force, and the ladder around it. Several instruments set a ladder of values rather than one: Brazil's CONAMA 506/2024 runs 20, 17, 15, 10, $5 \mu\text{g}/\text{m}^3$ across five stages, of which 17 has been in force since January 2025; the European limit is 25 with 10 from 2030; the United Kingdom's limit value is 20 with an English target of 10 by 2040; the United States moved from 12.0 to 9.0 in 2024; Bangladesh from 15 to 35 in 2022. The T1 key is the value in force in the collection window, and it is exacting: no model returns 17 for Brazil (the modal answer is 15, the next stage), almost none returns 20 for the United Kingdom (most return the 2040 target or the superseded 25), and fewer than a quarter return 9.0 for the United States. Of the 759 wrong T1 answers, 281 cite a value that sits somewhere on the country's official ladder, a stage not yet in force or a value since replaced, which is confusion about which value binds rather than fabrication, and is the failure

a training cut-off would produce. The distinction does not rescue the tier comparison; it sharpens it. Scoring any ladder value as correct raises Global North accuracy on T1 from 0.51 to 0.77 and Global South accuracy only from 0.31 to 0.42, because the staged instruments are mostly in the North, and the conditional odds ratio falls to 0.14 [0.11, 0.16]. Where the register offers several plausible numbers the models know them; where it offers one, they often do not.

4.10. Qualitative failure modes

Three failure modes recur behind the composite, with verbatim examples in Supplementary Material. Models fabricate standards that do not exist for the three countries with no national norm: in 168 of the 256 scored T1 cells for Angola, Argentina and Nigeria the answer asserts a specific national value, in 33 it correctly states that none exists, and in 31 it declines to answer, and the invented values disagree among replicates. They miss the binding value, answering 15 $\mu\text{g}/\text{m}^3$ for Brazil where the stage in force is 17 (CONAMA 506/2024), a confusion of stage rather than an invention. And they degrade in the native language, as when GPT-OSS 120B returned India’s correct 40 $\mu\text{g}/\text{m}^3$ in English and an empty answer in Hindi.

4.11. Inter-judge reliability

Reliability is reported on the sample that produces the effects, not on a calibration subset. All 3,190 items requiring a judge were scored by all three panel members, giving Krippendorff’s $\alpha = 0.527$, single-judge $\text{ICC}(2, 1) = 0.558$, and, as the operative quantity since the analysis uses the panel mean, $\text{ICC}(2, 3) = 0.791$.

The panel also isolates a design rule that generalises beyond this study. Where the judging prompt asked the model to compare a returned concentration against a tolerance band expressed in words, that is, to perform the arithmetic itself, inter-judge concordance on those items was $r = -0.04$. Pre-computing the admissible range and asking only whether the value falls inside it raised concordance on the same items to $r = +1.00$, and the 274 scores collected under the ambiguous wording were discarded and recollected. Apparent judge disagreement can be an artefact of what the judge is asked to compute, which is the same lesson [Zheng et al. \(2023\)](#) draw when a reference answer cuts a judge’s failure rate on arithmetic from 70% to 15%. Arithmetic belongs to code.

4.12. Robustness

The three supported effects were tested against seven attacks. The native-language penalty and the recall floor survive all of them; the tier gap survives five and is marginal under two, which the next paragraph states exactly. The development gradient stays below the pre-specified 0.55 under every one (its highest value, 0.48, is a leave-one-out re-estimate). Full outputs are in Supplementary Material.

No single country, and no weighting, carries the result. Leave-one-country-out re-estimation holds the composite tier gap in $[+4.4, +6.0]$ pp and the native-language penalty at -4.3 to -5.5 pp. Under equal weights the gap is $+4.2$ pp and the language penalty -4.8 pp; restricting to factual accuracy against the register value widens them to $+8.9$ and -6.1 pp and deepens the recall floor, so that both effects grow where the measurement is most objective. The two replicates of each cell differ by 8.5 pp on average

(within-cell SD 10.1 pp), so the -5.0 pp language penalty, averaged over 839 pairs, stands roughly ten standard errors from the noise two replicates leave unresolved.

Where the tier gap is established and where it is marginal. Specifications of one dataset are not independent tests, so we state each and what it assumes rather than counting them. The finding is the pooled code-adjudicated contrast, where no judge contributes and the country is the unit ($+14.7$ pp, $[+5.6, +23.9]$, permutation $p = 0.012$). On the composite, a permutation test over the 25 country means gives $p = 0.019$ and a Bayesian hierarchical model with crossed random intercepts agrees on sign and magnitude ($\beta = -0.049$, 94% HDI $[-0.088, -0.006]$, $P(\beta < 0) = 0.987$), while the frequentist mixed model, which is the specification that models the unbalanced design correctly, leaves it marginal ($\beta = -0.063$, $p = 0.086$) because it charges the large between-country heterogeneity to the error term, and the balanced-prompt subset gives $p = 0.32$. The gap is therefore supported on the code-only outcome and on the composite by permutation, and marginal under the mixed model and on the balanced subset. Its E-value is 1.71 at the point estimate and 1.31 at the confidence limit nearest the null ([Vander-Weele and Ding, 2017](#)): a country-level confounder of modest strength would suffice to explain it away.

The gap follows the UNCTAD partition, not the HDI partitions. Re-estimated under three development-based partitions of the same 25 countries, the gap is significant under UNCTAD and non-significant under every split drawn on HDI (Supplementary Material). This is consistent with the null HDI gradient, since a split on HDI should not then be expected to produce

a gap, and it concerns the accuracy level rather than the country-specificity deficit, where development is the exposure that does clear the threshold (Section 4.7). One reading of the first is that position in the world system rather than income separates the accuracy level (Quijano, 2000; Mohamed et al., 2020); the reading is post hoc, and we claim the gap for the North/South distinction as conventionally drawn, not as a general statement about development.

What the null gradient does and does not exclude. Simulating 25-country samples at known correlations, the design has 77% power at $\rho = 0.55$, the threshold fixed in advance, and a minimum detectable effect of $\rho = 0.56$. Power says how often a true gradient of that size would have been detected; it does not make the observed interval narrower, and the interval reported above still includes 0.55. The ten countries added post hoc fall within the HDI range the original fifteen already spanned, so the extension raises the density of observations rather than the range of the predictor.

The regional model was served as published. Because H3 rests on one locally served model, we verified the serving stack after the fact. The Ollama template used for Cabra-Mistral 7B is the chat template published with the model’s tokenizer, and the model answers the Portuguese prompts in coherent Portuguese. We then re-served the 30 Portuguese prompts, with two replicates, to Cabra, to its declared base (Mistral-7B-Instruct-v0.3 at the same quantisation) and to Llama 3.1 8B, and scored T1–T3 with the study’s code extractor (Supplementary Material). Re-served, Cabra returns the register value on 47% of the code-resolved Portuguese items (37% in the original collection), its base model on 50% and Llama 3.1 8B on 43%. On 30 items

per model the three are within a few answers of one another: the fine-tune does not outperform the base it was trained from, the serving stack does not explain its rank, and the composite margin against Llama 3.1 8B comes from the judged tasks and subcomponents rather than from the register values.

4.13. Summary

Table 7 maps each hypothesis to its finding, the candidate mechanism, and an explicit evidence status. Three effects are supported, two of them under every attack in Section 4.12 and the tier gap under five of seven; two pre-specified expectations are not met; and the pre-specified corpus mechanism gives way to a development interaction that the design cannot separate from coverage.

5. Discussion

We set out to measure whether LLMs answer air pollution policy questions as reliably for the Global South as for the Global North, whether prompt framing changes the answer, and what mechanism drives any gap. Across 25 countries, 14 models, four languages, five tasks and two personas, scored against official ground truth by code where the answer is a register value and by a three-vendor panel elsewhere, the harms we can demonstrate are practical: the three remedies an agency would reach for all fail, the most intuitive of them makes the answer materially worse, and models are least reliable on the recall of the binding value in force. A North/South tier difference persists, and the mechanism behind it is not established at the level where inference is sound.

5.1. *Three intuitive remedies, all failing, and one that harms*

The local language does not help and materially hurts. Asking in the country’s own language lowers accuracy by 5.0 pp, significantly in all three languages and with the country as the unit of inference, and the largest penalty falls on Hindi, the language with the scarcest corpus, with India, the highest-scoring country of all 25 in English, falling furthest. This is the direction the resource hierarchy predicts (Joshi et al., 2020; Xue et al., 2021), and the measurement can see it because the unit is the language rather than the country; with three languages, and Spanish and Portuguese indistinguishable, the ordering is descriptive. The effect depends on no single country, model, task, persona condition or scoring instrument, nor on translation quality, and it holds on an outcome involving no judge at all: the model becomes markedly less likely to produce a checkable number in the first place, and more likely to return nothing.

The local-manager persona is equivalent to no persona. The difference-in-differences is +0.5 pp (permutation $p = 0.29$) and its country-level interval sits inside a ± 2 pp margin, so this is an equivalence result rather than an underpowered null, consistent with evidence that expert personas do not improve factual accuracy (Zheng et al., 2024; Basil et al., 2025), and its main effect is a small loss (-0.62 pp). This null carries more weight than a null about a folk habit would, because role framing is the first thing every major vendor tells its users to do; Anthropic, OpenAI, Google and xAI converge on the same advice (Section 2), so that an administration following its supplier’s documentation writes the sentence we tested. What that documentation claims for the sentence is control over tone, format and framing, and nothing

in it says that the technique which shapes register does not also shape factual reliability. Our estimate separates the two. A manipulation check confirms that half the responses explicitly take up the role, so the frame is received and changes the voice; it does not change what the model knows. We tested it typed into the message rather than installed as a system instruction, the channel a public servant in a chat interface has (Section 5.7).

The regional model buys nothing over the model it was tuned from. On the register values, re-served on the same Portuguese items, Cabra returns the value in force 47% of the time and its base model 50%: instruction tuning on Brazilian Portuguese did not improve what the model recalls about Brazilian regulation. It also ranks last of the fourteen and trails its scale-matched peer on the Portuguese prompts it was tuned for (0.258 against 0.313 for Llama 3.1 8B; -5.9 pp across the 30 Portuguese prompts), but those comparisons run on the composite, whose judged subcomponents come from the collection judges, and the roster rank compares a 7B model with systems an order of magnitude larger. The defensible claim is the narrow one, and it is the one an agency needs: downloading a regionally tuned small model does not recover the national facts a larger model misses. Two limits bound it. The narrow test is 30 items per model, and the model we could run is a community fine-tune of a 7B base, not a commercial regional frontier model of the kind an agency might procure. For those the finding is a prediction, and the released protocol is the way to test it.

Practitioners deploy all three as fixes. One does active damage, one is equivalent to doing nothing, and one adds nothing to the model it started from. For Global South environmental agencies, no cheap prompt- or model-

level workaround substitutes for verifying the binding value against the official source.

5.2. The recall floor sits on the values that bind

The risk falls unevenly. A manager drafting prose about air quality policy is comparatively well served; a manager retrieving the regulatory cutoff a decision turns on is in the worst cell of the design (Section 4.3), and worse still if they ask in a lower-resource language. The floor is not an artefact of scoring: the factual verdict on both recall tasks is decided by code against the register, the contrast holds with every model and every country as the unit, and a third of the wrong answers on the national standard cite a value from the country’s own official ladder rather than an invented one. The models know the ladder and not which rung binds.

5.3. A tier difference we can show, and a gradient we cannot

The geographic finding is a difference between groups, not a slope. On the outcomes no judge touches, Global South countries return the register value +14.7 pp less often than Global North countries, with the country as the unit; on the composite the gap is +5.0 pp, the same size inside the extension wave and on the pre-specified fifteen, and unchanged when the four EU members count as one. It is concentrated: it scales with how much the answer depends on a fact specific to the country and is absent on the one task with no register value to get right (Section 4.6), although the per-task intervals with the country as the unit are wide and that decomposition is exploratory. The development gradient stays below the pre-specified threshold on every reading, and its interval, $[-0.04, +0.66]$, includes both zero and the

threshold: 25 countries neither establish a gradient nor rule one out. The country ordering shows why the two results differ. India tops all 25 countries despite being Global South, Canada and France sit low, so the disadvantage is a tier difference layered over country-specific idiosyncrasy that a rank correlation over 25 countries cannot see through.

5.4. *The mechanism, and why it is not established*

The pre-specified mechanism test fails, and part of the reason is a defect in the instrument. The proxy assigns the size of the Wikipedia edition in a country’s dominant official language, so that nine of our 25 countries receive the value of the English Wikipedia. The variable does not measure how much text exists in a country’s language; it measures whether the country has English as an official language, which tracks colonial history rather than corpus size. We report this because it bears on any study that has used or will use the same proxy.

Changing the unit of variation buys one result and refuses another. At the country level the deficit on country-specific tasks is predicted by development ($\beta = +0.036$, $p = 0.027$ clustered; $\rho = -0.40$, $p = 0.049$ across the 25 deficits) and not, at conventional levels, by encyclopedic coverage ($p = 0.099$; $\rho = -0.37$, $p = 0.069$). The two correlate at $\rho = 0.65$ and the design has no lever that moves one without the other, so this does not identify development as the operative channel; it says that the exposure which clears the threshold is the one the pre-specified mechanism was meant to rule out.

The result is also narrower than the gradient it superficially resembles. Development does not predict how accurately models answer about a country (Section 4.6); it predicts how much worse they answer the questions whose

answer is a fact about that country. The two are different outcomes, and only the second has a country-level association. What remains excluded is the language-corpus channel, as it must be for a contrast measured entirely in English.

5.5. The methodological lesson: the key matters more than the sample

The same scored cells support different substantive conclusions depending on the instrument used to score them, and the size of that difference is the lesson. Scoring two tasks against placeholder keys and delegating range arithmetic to a judge yielded a significant development gradient and a small language penalty; rebuilding the keys from official registers and moving the comparison into code halved the gradient and more than doubled the penalty (Sections 4.6 and 4.4). Moving the factual verdict on the third register-valued task from two collection judges to code changed its headline odds ratio by a third, because the judges had been split along the same line as the tiers. The key itself kept moving under scrutiny: the United Kingdom entry carried a 2040 target in place of the limit in force, and the Bangladesh entry carried the 2005 value after the 2022 Rules had raised it, so that a model citing the correct current standard for a Global South country was being scored wrong until an external reader caught it. The dataset never changed; only the instrument did. The field’s instinct when a geographic-bias effect proves fragile is to add countries, but the ground truth deserves the scrutiny first. A defective key does not merely add noise; it adds structured noise, because the defects concentrate wherever official texts are hardest to obtain or most recently revised, which is where the hypothesis lives. The released registry therefore carries a content hash (Supplementary Material) so that

any reader can tell which version of the key a number was computed against. The same lesson applies to inference: with the response as the unit, H4 had $p = 6 \times 10^{-10}$; with the country as the unit it has $p = 0.069$, and only the second describes what 25 countries can show.

5.6. *Why the domain matters*

Fine particulate matter is among the leading environmental risk factors for premature mortality, and the burden falls disproportionately on the Global South: the World Health Organization attributes 88,548 deaths in Brazil in 2021 to ambient air pollution (uncertainty interval 69,477 to 108,431) ([World Health Organization, 2026](#)), and short-term exposure continues to drive measurable mortality in Brazilian cities ([Requia et al., 2024](#)). When a manager delegates a normative lookup or an episode recommendation to an LLM, an inaccurate answer can propagate into a regulatory brief or an emergency response.

We do not claim the model is least reliable where consequences are greatest, because we did not measure it: disease burden is not among our country covariates, and the one country-level observation we have points the other way, since India carries one of the largest $\text{PM}_{2.5}$ mortality burdens in the world and is also the highest-scoring country in our sample. Whether reliability is misaligned with need is answerable by adding an age-standardised attributable-mortality covariate, but not by this design. The coloniality-of-knowledge framework ([Mohamed et al., 2020](#)) offers one interpretation of the pattern we did measure, in which LLMs are compressed summaries of corpora produced under uneven digital and publication access and the language penalty is a link from corpus asymmetry to knowledge asymmetry; the data

support the penalty and the tier difference, and the interpretation remains an interpretation.

5.7. Limitations and future opportunities

Each limit below bounds this design, and each names the study that would remove it. We list them in the order we would run them.

No retrieval arm. Every answer comes from the model’s parametric memory. A manager using a search-augmented assistant, or a retrieval system over the national gazette, faces a different instrument, and the gap and the language penalty may both change under it. The released prompts and registries allow a retrieval arm to be run on the same items, and it is the first study we would add.

Role framing was tested in the user turn, not the system prompt. The manager frame is prefixed to the message rather than installed as a system instruction, where three of the four vendor guides place it; Google’s admits either that slot or the start of the user prompt, which is the one we used. The choice follows the population we care about, public servants in the chat interface, who have no system prompt to configure, but our null bounds that user’s channel, not a developer’s. Whether a system-instruction persona recovers factual accuracy where a typed one does not is a two-cell experiment on the same items.

From panel-scored to gold-anchored. No human expert re-scored responses here. Three properties bound what that costs: where the answer is a register value the factual verdict is computed by code; where judgement is required the score is the mean of three vendors rather than one; and every reported effect is a between-group contrast, in which a judge’s systematic

severity cancels. Restricting to objective factual accuracy strengthens both headline effects, the opposite of what a permissive judge would produce. Two residues remain: the four non-factual subcomponents of a T1 cell are still the collection judge’s, which is why the code verdict and not the composite is the primary tier outcome, and the extractor’s manual audit covered T1 only. A blinded human-expert validation of a stratified sample would convert an official-source-anchored result into a gold-standard-anchored one. The sample is exported with the release (`code/analysis/export_human_validation.py`): 150 responses stratified by tier, language and task, three rating sheets carrying the prompt, the answer and the official key but not the model, the tier or the machine’s score, and a key file opened only after the sheets return.

From a within-country signal to a causal design. Within countries we remove country-level confounding without removing task-by-country confounding, and the signal does not reach significance at the country level. A design that varies country coverage while holding development fixed, or a longitudinal study tracking accuracy as a country’s encyclopedic footprint grows, would address the first problem; more countries would address the second.

From a null gradient to an adequately powered test. The tier difference is established while the gradient is not, at $n = 25$, and the gradient’s interval includes the threshold. A pre-registered replication at larger n , which the open dataset makes cheap, would settle whether the gradient is absent or underpowered.

From three native languages to the full multilingual matrix. The strongest result rests on three native languages, so its mechanism is reported descrip-

tively even though the effect is not; for one of the three, Hindi, the official instrument is itself published in English, so the penalty there is a penalty for the user’s language rather than for the language of the register. The translation-quality objection is settled empirically by the back-translation census reported above; what remains open is breadth, and scaling to more languages is the highest-value extension this study points to.

From one domain to a cross-domain program. All effects come from one high-stakes applied domain and one corpus family (Wikimedia), and we confine the mechanistic language accordingly. The same task suite and ground-truth-registry method transfer directly to other regulated, source-anchored domains such as water quality, food safety and vaccine policy.

Served-model stack as the unit of analysis. We evaluate models as served through fixed access stacks, which is what a public manager queries; a residual gap could in principle reflect the stack rather than the weights. The serving check on the regional model (Section 4.12) covers one stack; a controlled design with identical open weights across stacks would separate the two for all of them, which the per-call provenance we release enables.

5.8. What this study contributes

The three fixes practitioners reach for first fail together, each in its own way: the local language harms, the persona frame is equivalent to no frame, and regional tuning at 7B adds nothing to the base model it started from. The methodological result is that in this literature the ground-truth key, and the level at which inference is carried, matter more than the sample size; changing those two, without touching a single response, moved a significant development gradient to a null, doubled the language penalty, and turned an

established corpus mechanism into an open question. And the design separates what 25 countries can demonstrate from what they cannot, reporting the second set as open rather than as supported.

6. Conclusion

Large language models are becoming working infrastructure for air pollution policy, a domain where an inaccurate answer about a binding standard, a measured concentration, or an operational response carries direct public-health stakes in the Global South, where the PM_{2.5} mortality burden is heaviest ([GBD 2019 Risk Factors Collaborators, 2020](#); [Requia et al., 2024](#)). We measured geographic and persona-modulated bias in that domain with a 25-country, 14-model, four-language benchmark (English plus the native language of nine countries), scored against ground truth anchored to official primary sources, by code wherever the answer is a register value and by the mean of a three-vendor judge panel elsewhere.

None of the remedies an agency can try works, and the most intuitive one backfires. Asking in the country’s own language lowers accuracy by 5.0 percentage points, most sharply for Hindi, the language least represented in training corpora, and the penalty is not an artefact of unanswered prompts. A local-manager persona is equivalent to no persona within two percentage points. Regional instruction tuning at 7B returns the value in force no more often than the base model it was tuned from. Accuracy is lowest on the recall of a binding value, the task a manager can least afford to have wrong, and a third of the wrong answers cite a rung of the country’s own official ladder rather than an invented number. A Global North/South gap persists on the

outcomes no judge touches, where Global South countries return the register value +14.7 pp less often with the country as the unit. Its concentration on tasks whose answer is a national fact is a pattern of point estimates: every per-task interval at the country level includes 1. The development gradient does not reach significance and its interval includes both zero and the pre-specified threshold. On mechanism, what predicts the country-specificity deficit at the country level is development rather than encyclopedic coverage, and the two are too collinear in 25 countries to be told apart; the language-corpus channel is excluded.

For Global South environmental agencies the practical message is cautionary. LLM outputs on binding standards and measured data should be treated as drafts checked against the official gazette, and none of the cheap prompt- or model-level workarounds substitutes for that verification, least of all switching to the local language, which on this evidence makes the answer worse. The penalty falls on the languages under-served in training corpora (Joshi et al., 2020), hardest on Hindi; whether those languages are also spoken where the health burden of air pollution is greatest is not something this design measures, and the one country-level observation we have (India) points the other way (Section 5).

One methodological result generalises beyond the domain. Rebuilding two task keys from official registers and moving range comparison out of the judge and into code changed the substantive conclusions on the same responses, and carrying inference at the country rather than the response level changed the status of the mechanism test. When an effect in this literature proves fragile, the ground-truth key and the unit of inference deserve the au-

dit before the sample does, because a defective key adds structured noise that concentrates where official data are hardest to obtain, and a response-level p -value overstates what a handful of countries can show.

Protocol, prompts, official-source ground-truth registries, analysis code, and anonymised response data are released openly, with a one-command reproduction pipeline and a method-audit gate that recomputes every reported number from the raw data. The study also provides a reusable template for applied LLM-equity audits, with official-source ground truth, code adjudication wherever the answer is a checkable number, and end-to-end reproducibility, in the high-stakes public-sector domains where these systems are now being deployed.

Data and code availability

The full protocol, prompts, analysis code, ground-truth registries, and the anonymised LLM responses will be released on publication at github.com/Roverlucas/artigo-vies-analise-regional under MIT (code) and CC-BY-4.0 (data, prompts, documentation) licenses. The release includes the per-response records that every reported effect is computed from, so each number in this paper can be recomputed from the raw data rather than taken on trust. The analysis pipeline runs end-to-end via `python code/run_all.py -confirmatory`.

Study protocol

Hypotheses, sampling frame, outcome definition, decision thresholds and multiplicity correction were fixed in a version-controlled analysis plan before

confirmatory collection began. That plan was not deposited in a public registry, so readers cannot verify its timing independently of the repository history, and we do not claim pre-registration. Delivered scope also departed substantially from the planned design, which the plan itself made a condition for confirmatory status.

We therefore report the study in two layers. The first states what was specified in advance and what became of it: the three co-primary hypotheses and the secondary hypothesis are reported against their declared thresholds, including the three that failed and the one whose registered direction the data reverse. The second reports what the data suggested and the plan did not anticipate, labelled exploratory throughout, with intervals and without confirmatory language. Deviations from the plan are tabulated in the supplement.

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Author contributions

Following CRediT taxonomy:

- Lucas Rover (UTFPR): Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Visualization, Writing – Original Draft, Project Administration.

- Vitor de Melo Dominski (Faculdade Descomplica): Data Curation, Software, Formal Analysis.
- Prof. Anibal Tavares de Azevedo (Unicamp): Writing – Original Draft, Writing – Review & Editing.
- Dr. Eduardo Tadeu Bacalhau (UFPR): Validation, Supervision, Writing – Review & Editing.
- Dra. Yara de Souza Tadano (UTFPR): Validation, Supervision, Writing – Review & Editing.

Declaration of generative AI in the writing process

Two distinct uses must be separated here, because only one of them is what this declaration is about.

As an object and instrument of the study. Large language models are the subject of this research, and they also serve as one of its measuring instruments: GPT-5-mini and Claude Haiku 4.5 scored model responses as collection judges, Gemini 2.5 Pro, Claude Sonnet 4.6 and DeepSeek-V3 re-scored them as a three-vendor panel, and Claude Sonnet 4.6 produced the native-language renderings of the prompts. These uses are part of the method, are described in Section 3, and are why the study also reports inter-judge reliability and a back-translation audit of the translated prompts. Wherever the correct answer is a value in an official register, the verdict is computed by code rather than by a model.

In preparing the manuscript. During the preparation of this work the authors used Anthropic’s Claude (via Claude Code) to assist with drafting and editing the text and implementing analysis code; OpenAI’s Codex CLI

to audit the manuscript, supplement and code against their sources; and, through their APIs, Gemini 2.5 Pro, DeepSeek-V3 and GPT-5.2-pro as independent reviewers of the manuscript’s claims and prose, each run from a vendor other than the one that drafted the passage under review. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No language model is listed as an author, and no model contributed the interpretation of findings or the decisions about what the evidence supports.

Declaration of interests

The authors declare no competing interests.

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Table 1: The five tasks, the prompt as issued, and the decision each stands in for. Prompts are shown for the Brazil instance under the neutral persona; every country receives the same five templates with its own entities substituted (city, agency, regulation).

Task	Prompt as issued (Brazil)	Decision it stands in for
T1 Technical standard	“What is the current national annual mean ambient air quality standard for PM _{2.5} in Brazil? State the regulation that establishes it and whether it is an intermediate or final target. Limit to 1–3 sentences.”	Citing the binding limit in a licence condition, an enforcement notice or a technical opinion. The wrong value voids the act.
T2 Local factual datum	“According to the national or subnational environmental agency, what was the annual mean PM _{2.5} concentration in São Paulo, Brazil, in the most recent reported year? Provide value in $\mu\text{g}/\text{m}^3$, the reference station or network, and the source.”	Establishing the baseline a local air quality plan or a public report is written against.
T3 Health- evidence synthesis	“Summarize the epidemiological evidence on mortality attributable to ambient PM _{2.5} exposure in Brazil over the last decade. Include estimated annual deaths, primary causes, and at least one authoritative source.”	Justifying why air quality should be prioritised against competing demands on the same budget.
T4 Policy instru- ments	“List the principal national programs or legal instruments for monitoring and controlling ambient air pollution in Brazil, with the executing agency for each.”	Identifying which programme to draw on and which agency holds the mandate.
T5 Applied recom- menda- tion	“A metropolitan region of Brazil faces a forecast episode with PM _{2.5} expected to exceed $75 \mu\text{g}/\text{m}^3$ within 24 hours. Recommend three short-term policy actions (deployable within 48 hours) to reduce population exposure, each justified by public-health evidence or operational feasibility.”	Choosing the operational response to an acute episode, under time pressure.

Table 2: Composite accuracy by model across 25 countries (factual verdict by code where the answer is a register value, panel mean elsewhere; $[0, 1]$). N = scored English-prompt responses.

Model	N	Mean
DeepSeek-V3	496	0.687
GPT-5	496	0.674
Gemini 2.5 Flash	486	0.634
GPT-5-mini	495	0.626
Claude Haiku 4.5	486	0.591
Qwen3 32B	494	0.562
Command R+	496	0.530
Llama 3.3 70B	391	0.511
Llama 4 Scout	496	0.498
Qwen3 14B	496	0.485
GPT-OSS 120B	257	0.484
Phi-4 14B	496	0.477
Llama 3.1 8B	496	0.438
Cabra-Mistral 7B	492	0.351

Table 3: Composite accuracy by task type (25 countries; T1 on 24, Egypt having no key). On T1, T2 and T3 the factual-accuracy subcomponent is the code verdict where the returned value is checkable against the official register, and the remaining subcomponents are judge-scored; T4 is panel-scored; T5 retains the collection judge’s scores.

Task	N	Mean
T5 (applied recommendation)	1,320	0.769
T3 (health-evidence synth.)	1,329	0.548
T4 (policy instruments)	1,310	0.520
T2 (local factual datum)	1,335	0.473
T1 (technical standard)	1,279	0.393

Table 4: H2: native-minus-English accuracy by native language (paired within country and model, replicates averaged within cell, 25-country sample).

Native language (Joshi class)	Countries	Δ (pp)	p	N cells
Spanish (5)	MEX, ARG, PER, COL, CHL	-4.5	2×10^{-10}	457
Portuguese (4)	BRA, PRT, AGO	-4.2	8×10^{-5}	311
Hindi (4)	IND	-11.4	4×10^{-4}	71
Overall		-5.0	5×10^{-16}	839

Table 5: Global South versus Global North odds of a factually correct answer, by task (exploratory decomposition). The outcome is factual accuracy ≥ 0.5 : the code verdict where the task is adjudicated by code (all of T1; the resolvable part of T2 and T3), the panel or judge score otherwise, so for T5, which has no register value, it is the judge’s factual rating. Conditional OR from a binomial mixed model with crossed random intercepts for country and model, an interval for a within-country, within-model contrast; raw OR with its 95% country-level bootstrap interval and the permutation p of the tier label with the country as the unit. The two differ by roughly a factor of five in width because the country intercepts are far from exchangeable (variance components in Supplementary Material). Every country-level interval includes 1; the decomposition is exploratory and is read as a pattern of point estimates.

Task (answer is)	GN	GS	OR cond.	OR raw [95% country CI]	p
T1 standard (national value)	50.7%	30.9%	0.32	0.44 [0.18, 1.05]	0.062
T4 instruments (national)	62.0%	44.7%	0.33	0.50 [0.19, 1.15]	0.156
T2 local datum (national value)	50.7%	37.8%	0.47	0.59 [0.34, 1.00]	0.086
T3 health synthesis (international)	59.1%	52.3%	0.69	0.76 [0.37, 1.46]	0.397
T5 recommendation (no register value)	99.6%	99.7%	at ceiling in both tiers; no contrast estimable		

Table 6: Country coverage $\times$ task dependence, response-level model with a fixed effect for country. Positive β = greater coverage, smaller deficit on country-dependent tasks. p from the response-level mixed model (random intercept per model) and, for the primary rows, with standard errors clustered by country.

Task classification	β	p (response)	p (clustered)
T1, T2, T4 versus T3, T5 (primary)	+0.031	6×10^{-10}	0.099
T1 versus T5 (purest contrast)	+0.046	1×10^{-11}	
Excluding T5 (ceiling at 99.7%)	+0.026	3×10^{-5}	
Excluding T4 (most interpretive)	+0.034	6×10^{-10}	
Placebo: labels inverted	-0.031	6×10^{-10}	
HDI $\times$ dependence alone	+0.036	8×10^{-13}	0.027
With HDI $\times$ dependence: coverage	+0.013	0.057	0.69
HDI	+0.027	5×10^{-5}	0.35

Table 7: Hypotheses, findings, candidate mechanisms, and evidence status. Grouped by what the evidence supports, not by hypothesis number.

Hypothesis	Finding	Mechanism
<i>Supported</i>		
H2 language	−5.0 pp; Hindi −11.4	language corpus
H3 regional model	no gain over its own base (47% vs 50%)	tuning adds nothing at 7B
H1 tier gap	+14.7 pp code-only, country as unit	tier
<i>Established at country level, exposure not separable</i>		
H4 within country	HDI×dependence +0.036 ($p = 0.027$); coverage $p = 0.099$	development or coverage
<i>Not supported</i>		
H1 gradient	$\rho = 0.36 [-0.04, +0.66]$	development
H4 pre-specified	language-corpus proxy null	language corpus
H6 persona	DiD +0.5 pp, equivalent within ± 2 pp	—
<i>Descriptive only</i>		
H5 open/closed	+12.8 pp for closed	access type

Note. H2 is significant with the country as the unit ($p = 0.002$) and marginal with the model as the unit ($p = 0.058$, 12/14 negative); its mechanism rests on three native languages and is descriptive. H1’s composite gap is marginal under the mixed model and on the balanced-prompt subset; the per-task decomposition is exploratory and every country-level interval includes 1. H3 is the narrow contrast against the base model; the roster rank compares 7B with frontier systems. H4’s development interaction predicts the country-specificity deficit, not the accuracy level.